

Container Registry

Docker Hub · ECR · GCR · GHCR · Private Registries

■ Analogy

- A container registry is like a library for Docker images. Docker Hub is the public library — anyone can borrow books. ECR/GCR are private company libraries — only employees with a badge can access. Tagging an image is like labeling a book edition — `myapp:1.2.0`, `myapp:latest`.

Key Terms

Registry	Server storing Docker image layers and manifests. Default: Docker Hub (docker.io)
Repository	Collection of related images with different tags — e.g., <code>mycompany/myapp</code>
Tag	Label on an image: <code>myapp:1.0.0</code> , <code>myapp:latest</code> . 'latest' is just a convention — not magic
Docker Hub	Public registry at <code>hub.docker.com</code> . Free tier: 1 private repo, rate-limited pulls
ECR	AWS Elastic Container Registry. IAM-authenticated, integrates with ECS/EKS
GCR / Artifact Reg	Google Container Registry (legacy) / Artifact Registry (new). GKE-native
GHCR	GitHub Container Registry — <code>ghcr.io</code> . Free for public repos, integrates with GitHub Actions
docker login	Authenticates CLI to a registry. Stores credentials in <code>~/.docker/config.json</code>
Image manifest	JSON describing image layers and config. Enables multi-arch images (manifest list)
Multi-arch image	Single image tag that works on amd64 and arm64 — built with <code>docker buildx</code>

Registry Comparison

Registry	Auth	Best For	Free Tier
Docker Hub	Username/PAT	Public OSS, quick testing	1 private repo
AWS ECR	IAM role/OIDC	ECS/EKS workloads	500MB free
GCR / Artifact Registry	GCP SA/Workload Identity	GKE workloads	Pay per GB
GHCR	GitHub token/OIDC	GitHub Actions CI/CD	Free for public
Harbor (self-hosted)	LDAP/OIDC	Air-gapped, compliance	Free (OSS)

JFrog Artifactory	LDAP/SSO	Enterprise multi-format	Free tier
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Push to ECR via GitHub Actions

```
# In GitHub Actions workflow:
- name: Configure AWS credentials
  uses: aws-actions/configure-aws-credentials@v4
  with:
    role-to-assume: arn:aws:iam::123456789:role/github-actions
    aws-region: us-east-1
- name: Login to ECR
  uses: aws-actions/amazon-ecr-login@v2
- name: Build and push
  run: |
    IMAGE=123456789.dkr.ecr.us-east-1.amazonaws.com/myapp
    docker build -t $IMAGE:${{ github.sha }} .
    docker tag $IMAGE:${{ github.sha }} $IMAGE:latest
    docker push $IMAGE:${{ github.sha }}
    docker push $IMAGE:latest
```

Build multi-arch image with buildx

```
# Create a multi-arch builder
docker buildx create --use --name multiarch

# Build and push for amd64 + arm64 in one command
docker buildx build \
  --platform linux/amd64,linux/arm64 \
  -t myrepo/myapp:1.0.0 \
  --push .
```

■ Real-World: Pinterest

- Pinterest manages 1000+ microservice images in AWS ECR. They use lifecycle policies to auto-delete untagged images older than 7 days, OIDC authentication from GitHub Actions (no static AWS keys), and image scanning on push to catch CVEs before deployment.